

Analysis First-Year Exam, January 2017, Part 1

Answer *FOUR* questions in detail. State carefully any results used without proof.

1. For each λ in the nonempty index set Λ , let X_λ be a nonempty subset of $\mathbb{R}$ and assume that the union $X = \bigcup\{X_\lambda : \lambda \in \Lambda\}$ is bounded above. Is $\sup\{\sup X_\lambda : \lambda \in \Lambda\}$ defined? Does it equal $\sup X$? Prove.
2. Let A be a subset of the metric space M and for $\delta > 0$ define

$$A_\delta = \{x \in M : (\exists a \in A) d(x, a) < \delta\}.$$

Prove

- (i) that A_δ is open
 - (ii) that $\bigcap_{\delta>0} A_\delta = \overline{A}$.
3. Let $K_0 \supseteq K_1 \supseteq \dots$ be a decreasing sequence of nonempty subsets of the metric space M .
 - (i) Prove that if each K_n is compact then $\bigcap_{n \geq 0} K_n$ is nonempty.
 - (ii) Show by example that if each K_n is closed then $\bigcap_{n \geq 0} K_n$ can be empty.
 4. Let $f : X \rightarrow Y$ be a continuous bijection, with X complete. Prove that if f^{-1} is uniformly continuous then Y is complete. Give an example to show that dropping uniformly can jeopardize this conclusion.
 5. For a differentiable function $f : (0, \infty) \rightarrow \mathbb{R}$ consider the implications: One is true and the other is false; prove the true and give a counterexample for the false.
 - (i) $\lim_{t \rightarrow \infty} f'(t) = 0 \Rightarrow \lim_{t \rightarrow \infty} (f(t)/t) = 0$;
 - (ii) $\lim_{t \rightarrow \infty} (f(t)/t) = 0 \Rightarrow \lim_{t \rightarrow \infty} f'(t) = 0$.